



# Beyond Keywords: Building an Intent-Driven Spatial Search Engine

Mapping human-centric vocabulary to geographic realities.

Transitioning from what a place is named to what a user actually needs.

# The Problem: Local Intent Failure

## The Quiet Place

### Paradox

When a user in Pilani searches for a "quiet place," the system returns a business named "Quiet Place" located 1,000 km away in Bengaluru. The search prioritises exact business name matches over geographic proximity and user intent.

## Exact-Match Bias

Current systems fundamentally misunderstand the query—they're searching for places *named* "quiet place" rather than places that *are* quiet.



# The Indian Context & Research Gaps

Structured street addresses are insufficient in India, where navigation is descriptive and informal. Users search by landmarks, local nicknames, and contextual descriptions rather than formal addresses.

## Gap 1: Exact-Match Bias

Systems prioritise business titles over geographic proximity and semantic meaning

## Gap 2: Deterministic Filtering

Rigid rules fail to capture the fuzzy, contextual nature of human spatial reasoning

## Gap 3: Decoupled Architecture

Natural language processing remains disconnected from the spatial database layer





# The Proposed Solution:

## Intent-Based Spatial Routing

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### Conversational Query

User asks: "Find a quiet place near me"

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### Semantic Ontology

Maps "quiet" to formal geographic tags

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### Multi-Dimensional Filters

Structured spatial and attribute queries

4

### Geographic Results

Location-aware recommendations

Bridge the gap using a **Semantic Ontology** that maps human needs (e.g., "vibe", "utility") to formal geographic tags.

# Technical Architecture & Model Selection

## From Cloud to Local

Transition from API-dependent models like Llama-3-70B to a locally deployed, specialised model: **Defog**

**SQLCoder-8b.**

## Why Local Deployment?

- Enhanced data privacy and security
- Zero network latency for real-time queries
- Deep integration with PostGIS for spatial operations
- Custom optimisation for Indian spatial patterns





# The Data Foundation

## The Database



A denormalised PostGIS flat table (pilani\_places) sourced from OpenStreetMap, containing enriched spatial and attribute data

## The Ontology



Human intents like "Late Night" or "Study" map to specific SQL ILIKE patterns or boolean flags for efficient querying

The ontology bridges conversational language with formal database queries, translating "I need somewhere to study late at night" into structured filters.



# Research Foundations & Academic Baselines

Our methodology is built upon critical academic precedents, addressing their limitations to achieve a more robust evaluation.



## Spider: Complex Semantic Parsing

Yu et al., 2018

Introduced "Exact Set Match" for SQL structure. Its limitation: generates **False Negatives** by penalising valid SQL that differs structurally from the gold standard.



## Test Suite Accuracy

Zhong et al., 2020

Emphasised execution-based validation against multiple database states. Its limitation: in denormalised databases, it can lead to **False Positives** where a lucky result masks a logically flawed query.

By merging the structural rigor of the Spider approach with the empirical validity of the Test Suite approach, our Hybrid Framework ensures a query is both logically sound and empirically accurate before being marked as "PASS".

# The Hybrid Evaluation Framework

## Why Traditional Metrics Fail

False positives in execution versus false negatives in structural matching require a multi-dimensional evaluation approach.



### AST-Based Structural Testing

Using sqlglot to validate SQL query structure and syntax correctness



### Execution-Based Semantic Testing

Running queries against the database to verify meaningful results



### Hybrid Scoring

Combining structural and semantic metrics for comprehensive evaluation



# Evaluation Methodology & Dataset Testing

Our robust evaluation methodology combines a custom benchmark with a hybrid testing framework to accurately assess model performance in the complex Indian spatial context.

## Benchmark Design



### Custom Golden Benchmark

50 manually authored spatial queries reflecting real-world Indian intents (e.g., "late-night cafes," "places that accept UPI").



### Complexity Stratification

Queries categorized into four difficulty levels: Easy, Medium, Hard, and Extra Hard to assess analytical depth.

## Hybrid Evaluation Framework

### Dual Testing Pipeline

Eliminates high false positives/negatives common in traditional academic benchmarks.

### AST-Based Structural Testing

Uses `sqlglot` to verify correct table, column, and operator usage, preventing schema hallucination.

### Execution-Based Semantic Testing

Predicted queries are run against a live PostGIS database via `SQLAlchemy` for meaningful result verification.

### F1-Score Validation

A threshold of  $\geq 0.50$  F1-Score ensures core spatial intent capture despite real-world data entropy.

### Dynamic Constraint Stripping

Arbitrary constraints (e.g., `LIMIT 5`) are removed for a fair comparison of entire result sets.

# The 'F1-Score': Mathematical Justification for Spatial Accuracy

Traditional "Pass/Fail" metrics fall short in real-world geographic data, where noise and minor inconsistencies are common. We need a nuanced approach to evaluate spatial intent.



## Precision

Measures how many returned locations were correct, avoiding "hallucinated" results.



## Recall

Ensures completeness, measuring how many "Gold" locations the model successfully found.



## F1-Score

The harmonic mean of Precision and Recall, penalising both vague (low precision) and overly restrictive (low recall) query generation.

A 0.50 F1 threshold validates that the core spatial intent is captured, accounting for spatial data's inherent entropy and allowing for "partial matches." Our system achieved an average F1 Semantic Score of **0.784**, demonstrating empirical reliability despite messy colloquial inputs.

# Experimental Results

**84%**

**Overall Pass  
Rate**

Successful query  
generation across all  
complexity levels

**0.827**

**Average  
Structural Score**

Syntax and query  
structure correctness

**0.784**

**Average  
Semantic  
F1-Score**

Precision and recall  
of meaningful results





# Performance by Complexity



90%

## Easy & Medium Queries

Simple intent matching with high accuracy



68%

## Hard & Extra Hard Queries

Complex logic and negations challenge the system

## Performance Drop Factors

System struggles with asymmetrical tagging patterns and strict negation handling in complex queries.



# Future Work & Conclusion

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## Expand Semantic Ontology

Enhanced entity resolution for local nicknames and informal place references

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## Build Custom Chatbot

Move from generic LLaMA prototype to specialised conversational interface

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## Improve Spatial State Management

Better context handling in follow-up questions with data-to-dialogue synthesis

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## Address String Matching Fragility

Handle variations like "Meera Bhawan" versus "Meera Bhawan girls hostel"

**Final Goal:** A scalable, intuitive navigational framework reflecting how people actually perceive their surroundings, bridging the gap between human-centric vocabulary and geographic realities.